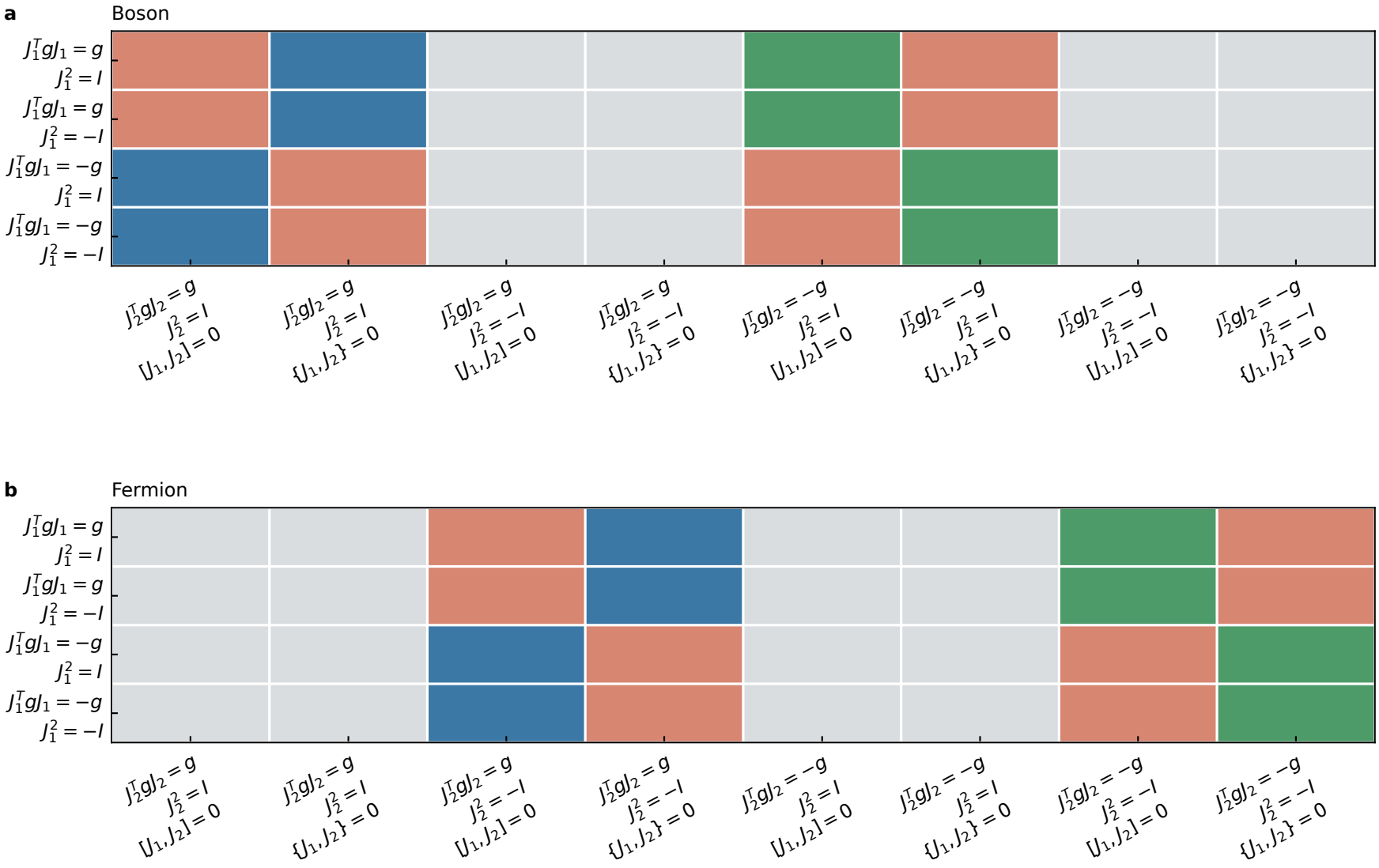
a
Boson
$J_1^T g J_1 = g$
$J_1^2 = I$
$J_1^T g J_1 = g$
$J_1^2 = -I$
$J_1^T g J_1 = -g$
$J_1^2 = I$
$J_1^T g J_1 = -g$
$J_1^2 = -I$
$J_2^T g J_2 = g$ $J_2^2 = I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = g$ $J_2^2 = I$ $\{J_1, J_2\} = 0$
$J_2^T g J_2 = g$ $J_2^2 = -I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = g$ $J_2^2 = -I$ $\{J_1, J_2\} = 0$
$J_2^T g J_2 = -g$ $J_2^2 = I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = -g$ $J_2^2 = I$ $\{J_1, J_2\} = 0$
$J_2^T g J_2 = -g$ $J_2^2 = -I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = -g$ $J_2^2 = -I$ $\{J_1, J_2\} = 0$
b
Fermion
$J_1^T g J_1 = g$
$J_1^2 = I$
$J_1^T g J_1 = g$
$J_1^2 = -I$
$J_1^T g J_1 = -g$
$J_1^2 = I$
$J_1^T g J_1 = -g$
$J_1^2 = -I$
$J_2^T g J_2 = g$ $J_2^2 = I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = g$ $J_2^2 = I$ $\{J_1, J_2\} = 0$
$J_2^T g J_2 = g$ $J_2^2 = -I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = g$ $J_2^2 = -I$ $\{J_1, J_2\} = 0$
$J_2^T g J_2 = -g$ $J_2^2 = I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = -g$ $J_2^2 = I$ $\{J_1, J_2\} = 0$
$J_2^T g J_2 = -g$ $J_2^2 = -I$ $[J_1, J_2] = 0$
$J_2^T g J_2 = -g$ $J_2^2 = -I$ $\{J_1, J_2\} = 0$
$\eta A + A^\dagger \eta < 0,\ \eta = J_2 g$
$\eta A + A^\dagger \eta < 0,\ \eta = i J_2 g$
$J_1^T \overline{(\eta A + A^\dagger \eta)} J_1 = -(\eta A + A^\dagger \eta)$
$\mathrm{Tr}(\eta A + A^\dagger \eta) = 0$